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CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Of course. As STRUCTMIND AI, I will provide a detailed connection design advisory for your 3-storey SMF + OCBF mixed system.

This analysis assumes the lateral force-resisting system (LFRS) is clearly delineated, with moment frames and braced frames in separate, defined lines. All seismic provisions reference AISC 341-22, and strength provisions reference AISC 360-22.

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1. Beam-to-Column Joints (Special Moment Frame - SMF)

These connections are required to be highly ductile and must be prequalified per AISC 341-22 Chapter E.

• 1. Recommended Connection Type: Reduced Beam Section (RBS) or "Dogbone"

This is often the most economical and widely used prequalified SMF connection. It strategically weakens the beam to force a ductile plastic hinge to form away from the column face, protecting the column and the critical beam flange groove welds.

• 2. Governing Limit States (per AISC 360-22 & 341-22):

The design philosophy is to ensure the connection is stronger than the yielding beam. Beam: Flexural Yielding (Plastic Hinge Formation) at the RBS location is the intended behavior, not a failure limit state. Checks for flange and web local buckling (AISC 360, Chapter F) are critical. Column: Panel Zone Shear Yielding (AISC 341, E3.6g) is often the governing limit state for the column, potentially requiring doubler plates or thicker column webs. Column: Flexural strength of the column at the joint must satisfy the "Strong Column-Weak Beam" requirement (AISC 341, E3.4a). Welds: The beam flange-to-column flange welds are Complete Joint Penetration (CJP) groove welds, classified as "Demand Critical" per AISC 341. Their strength is governed by the strength of the base metal. Shear Tab & Bolts: Bolt Shear Rupture (AISC 360, J3.6) Bearing/Tearout on Beam Web & Shear Tab (AISC 360, J3.10) Block Shear Rupture on Beam Web (AISC 360, J4.3)

• 3. Bolt / Weld Sizing Guidance:

• Welds: Use CJP groove welds for beam flanges to the column flange. These must conform to AWS D1.8 (Seismic Supplement). The shear tab is connected to the column with fillet welds or CJP welds.

• Bolts: High-strength ASTM F3125 Gr. A325 or A490 (or their twist-off equivalents). Size for the shear resulting from $1.1R_yM_p/L_h$ (where L_h is distance between plastic hinges) plus gravity shear.

Bearing-type connections are typical.

• 4. Constructability Score: 8/10

• Remarks: Widely understood by fabricators and erectors. Fabrication requires precise CNC cutting of the RBS profile. Field erection is straightforward: bolt the web tab, then perform the CJP flange welds. Requires qualified welders and NDT (UT) inspection.

• 5. Alternate Solutions:

• Bolted Flange Plate (BFP): Excellent alternative that minimizes field welding. Can be faster to erect but requires more material (plates) and bolts.

• Bolted Extended End-Plate (BEEP): Very fast erection, but requires tight fabrication tolerances on the column (plumbness and squareness) and can lead to very thick end plates and large diameter bolts.

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2. Column Base Plates

The design will differ significantly between SMF columns (moment + axial + shear) and OCBF columns (high axial + shear).

• 1. Recommended Connection Type: Exposed Column Base Plate

A thick steel plate is welded to the bottom of the column and anchored to the concrete foundation with cast-in-place anchor rods. For SMF columns, a larger anchor bolt pattern is used to create a moment-resisting couple.

• 2. Governing Limit States (per AISC 360-22, J8 & ACI 318-19, Ch. 17):

This is a concrete-steel interface design, so both specifications are critical. Steel Side (AISC 360): Base Plate Bending at critical sections around the column profile. Weld strength (column-to-base plate). Concrete/Anchor Side (ACI 318): Concrete Bearing Strength under the base plate. Anchor Rod Tensile and Shear Strength. Concrete Breakout Strength in Tension (often governs for uplift). Concrete Breakout Strength in Shear. Concrete Pryout Strength in Shear. Side-Face Blowout Strength (for anchors near an edge).

• 3. Bolt / Weld Sizing Guidance:

• Anchor Rods: Use ASTM F1554 Gr. 36, 55, or 105. Sizing is an iterative process governed by the ACI 318 concrete breakout and pryout checks. For SMF bases, ductility is critical.

• Welds: Typically fillet welds all around the column profile. For high-moment SMF bases, CJP or PJP groove welds may be necessary at the column flanges to transfer moment effectively.

• 4. Constructability Score: 7/10

• Remarks: Success is highly dependent on anchor rod placement accuracy. Using a survey-verified template is non-negotiable. Requires careful planning for grout placement (e.g., grout holes in the base plate, sufficient standoff).

• 5. Alternate Solutions:

• Embedded Column Base: The column is embedded directly into the foundation/pier. Provides excellent fixity but complicates formwork and concrete placement significantly.

• Shear Lugs: If shear forces are too high for anchor rods and friction alone, steel lugs can be welded to the bottom of the base plate to bear against the concrete.

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3. Brace-to-Gusset Connections (Ordinary Concentrically Braced Frame - OCBF)

These connections must accommodate brace buckling while providing required strength. The Uniform Force Method (UFM) is the standard design procedure.

• 1. Recommended Connection Type: Bolted HSS Brace to Welded Gusset Plate

The gusset plate is shop-welded to the beam and/or column. The HSS brace member is shop-welded to end-tee or knife plates, which are then field-bolted to the gusset plate.

• 2. Governing Limit States (per AISC 360-22, J4 & AISC 341-22, F1):

The connection must be designed for the lesser of the expected brace tensile strength ($R_y F_y A_g$) and the maximum force the system can deliver. In compression, it must resist 1.1 times the expected brace buckling strength. Brace Member: Tensile Yielding (AISC 360, J4.1a) on the gross section. Tensile Rupture (AISC 360, J4.1b) on the net section. Block Shear Rupture (AISC 360, J4.3). Gusset Plate: Tensile Yielding on Whitmore Section (AISC 360, J4.1a). Compressive Buckling of the Gusset Plate (AISC 360, J4.4). Block Shear Rupture (AISC 360, J4.3). Bolts: Bolt Shear Rupture (AISC 360, J3.6). Welds: Strength of welds connecting the gusset to the frame.

• 3. Bolt / Weld Sizing Guidance:

• Bolts: High-strength bolts in a bearing-type connection are standard. Sizing is governed by the required strength from AISC 341.

• Welds: Use fillet welds to connect the gusset to the beam/column flanges/web. Size to develop the force calculated from the UFM.

• 4. Constructability Score: 8/10

• Remarks: A very common and well-understood connection. The critical detail is providing a clearance of at least two times the gusset thickness (2t) between the end of the brace and the beam/column. This creates a "folding line" allowing the gusset to deform during brace buckling without damaging the main frame members.

• 5. Alternate Solutions:

• Fully Welded Connection: The brace can be directly welded to the gusset plate. This is less common as it removes a point of field adjustment and makes replacement more difficult.

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4. Splice Connections (Column & Beam)

Splices are used to connect shorter, transportable pieces into their full design length.

• 1. Recommended Connection Type: Bolted Flange and Web Plate Splice

This is the industry standard for field splices due to its speed, reliability, and avoidance of difficult field welding.

• 2. Governing Limit States (per AISC 360-22, J4):

• Splice Plates:

• Gross Section Yielding (AISC 360, J4.1a).

• Net Section Rupture (AISC 360, J4.1b).

• Bolts:

• Shear Rupture (AISC 360, J3.6).

• Bearing/Tearout on plates and main member (AISC 360, J3.10).

• Main Member:

• Block Shear Rupture (AISC 360, J4.3) at the bolt group.

• Seismic Considerations (AISC 341):

• Column Splices: Must develop at least 50% of the flexural strength of the smaller member. Typically placed ~4 ft above the floor for access.

• SMF Beam Splices: Not permitted within the plastic hinge zone. Must be designed to develop the expected flexural strength ($M_{pe} = R_y F_y Z_x$) of the beam section.

• 3. Bolt / Weld Sizing Guidance:

• Bolts: High-strength bolts. For column splices in moment frames, slip-critical connections are often specified to minimize drift and ensure proper load path. For beam and OCBF column splices, bearing-type is usually sufficient.

• 4. Constructability Score: 9/10

• Remarks: The preferred method for field splices. Erection aids like guide lugs or tabs are highly recommended for aligning heavy column sections before bolting. Location should be coordinated for clear access.

• 5. Alternate Solutions:

• Complete Joint Penetration (CJP) Welded Splice: Provides a continuous member profile but is very expensive, slow, and difficult to perform correctly in the field. Requires extensive NDT and highly skilled labor. Generally avoided unless architecturally required.